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To cite this article: Yuxiang Zhao, Peishi Yu, Yin Tao, Xin Zhang, Maoyang Li, Wen Xu & Junhua Zhao (2025) Long-term stability and durability of direct-ink-writing 3D-printed sensors: challenges, strategies and prospects, *Virtual and Physical Prototyping*, 20:1, e2460211, DOI: [10.1080/17452759.2025.2460211](https://doi.org/10.1080/17452759.2025.2460211)

To link to this article: <https://doi.org/10.1080/17452759.2025.2460211>



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Published online: 16 Feb 2025.



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Long-term stability and durability of direct-ink-writing 3D-printed sensors: challenges, strategies and prospects

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ABSTRACT

The direct-ink-writing (DIW) 3D-printed sensors have shown tremendous potentials in various fields, owing to their outstanding performances and simplified manufacturing process. Most of the existing studies have mainly focused on achieving higher stretchability and superior sensitivity, while few works devoted to obtain the long-term stability and durability. As a result, although various DIW sensors exhibit distinguished functions, there are still lack of successful applications in practical engineering. Accordingly, this review seeks to bring the challenges of long-term stability and durability to the forefront in this field. Firstly, the influential factors and respective mechanisms for both challenges are explored. Subsequently, potential strategies were analysed to enhance the long-term stability and durability for DIW sensors, including the ink innovation, sensor-structural design and printing process optimisation. Finally, the urgent requirements, interdisciplinary route and long-term applications are prospected. This paper provides insight for the DIW sensors in addressing the challenges of long-term stability and durability for promoting them to real engineering applications.

ARTICLE HISTORY

Received 29 November 2024
Accepted 25 January 2025

KEYWORDS

Direct-ink-writing (DIW);
sensors; long-term stability;
durability

1. Introduction

The direct-ink-writing (DIW) technology, as an extrusion-based 3D printing technique, has attracted considerable interest in recent years due to its remarkable versatility in fabricating intricate structures with a diverse range of materials [1–3]. In contrast to conventional manufacturing techniques, DIW technology facilitates the meticulous deposition of viscoelastic inks, enabling the construction of bespoke geometries in a layer-by-layer manner [4, 5]. This approach allows for the expedited prototyping and production of functional components. Advancements in DIW technology and specialised inks have driven the development of versatile sensors tailored to specific mechanical, electrical, and environmental requirements [6, 7]. Researchers have extensively investigated a diverse array of sensor types and functionalities, including pressure sensors [8], strain gauges [9], temperature detectors [10], chemical analyzers [11], and biosensors [12]. These sensors are meticulously designed to address various operational requirements, leveraging innovative materials such as conductive polymers, nanocomposites, and hydrogels to achieve enhanced sensitivity, stability, and overall

performance. The capability of DIW to fabricate highly customised and multifunctional sensors underscores its transformative potential for applications spanning biomedical engineering [12], environmental monitoring, wearable electronics [13] and industrial automation.

Despite the significant advantages demonstrated by DIW sensors, their application in real-world industries and engineering remains relatively limited. The most existing research predominantly emphasises the high stretchability and superior sensitivity of DIW sensors. Stretchability refers to the ability of a sensor to withstand and maintain functionality under mechanical deformation [14], such as stretching or bending. Sensitivity, on the other hand, describes the sensor's responsiveness to external stimuli (e.g. strain or pressure) and is quantified as the change in signal output per unit of applied stimulus [15]. While exceptional stretchability and sensitivity are often regarded as critical performance metrics for evaluating 3D-printed sensors [16, 17], this focus tends to understate the importance of their long-term stability and durability. The long-term stability and durability are fundamental in determining whether 3D-printed sensors can be extensively and sustainably applied in

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practical scenarios. Long-term stability refers to the ability of a sensor to maintain consistent performance over time, whereas durability pertains to structural integrity throughout its service life. In real-world applications, environmental factors and fatigue loading pose significant challenges to sensor performance. Aging and degradation induced by environmental conditions can alter the sensor's responsiveness, while complex loading scenarios, such as fatigue and impact, can significantly reduce its operational lifespan [18]. Therefore, comprehensive research and analysis into the long-term stability and durability of DIW sensors are imperative. Such efforts are essential to establish a solid foundation for their widespread and sustained adoption in practical engineering and industrial applications.

This review aims to provide an overview of the current state for DIW 3D printed sensors, focusing on the long-term stability and durability challenges of DIW sensors. By exploring both challenges in depth, insights are provided on subsequent long-term applications of the sensors. Subsequently, this review also discusses some potential strategies to improve the long-term stability and durability of DIW sensors. In addition, we introduced some potential future research prospects that will further contribute to the prolonged and widespread application of DIW 3D printed sensors under various and complex conditions.

2. The advancement of DIW-printed sensors

DIW (Direct-ink-writing) is an extrusion-based layer-by-layer printing technique that involves pressure-driven deposition of a viscoelastic ink through a fine nozzle, as shown in Figure 1 (a). The pioneer works on DIW technology originated from the 1990s. Hartmann et al. introduced the concept of robot-assisted shape deposition

in 1994 [19]. In 1997, Cesarano and Calvert put forth a novel approach to free moulding, utilising a low binder slurry, which came to be known as Robocasting [20]. Subsequently, Lewis et al. further developed the 3D direct writing technique in 2002 [21] and 2004 [22], wherein they demonstrated the fabrication of functional materials via DIW technology. These initial studies established the groundwork for the subsequent advancement of DIW technology. The subsequent work of Lewis et al. (2006) [23] demonstrated the potential of DIW technology for 3D functional material printing. At this juncture, DIW technology began to gain traction in the public consciousness and experienced a period of accelerated development over the subsequent decade, as illustrated in Figure 1 (b).

2.1 Development of DIW technology

In the early research stage, DIW technology was mainly used to fabricate complex ceramic structures [25]. The major challenge for DIW technology of that time was to optimise the rheological properties of the inks [1] to satisfy the printing requirements for different materials. As time progressed, researchers gradually developed a wide range of pastes and inks for DIW 3D printing technology, including polymers [26–29], ceramics [30–33], metals [34–38], and glass [39–41]. In addition to the enhancement and optimisation of conventional inks, DIW technology has developed a series of new functional inks such as shape memory polymers [42–44], magnetic soft materials [45, 46], liquid crystal elastomers (LCEs) [47–49], metal wires [50], conductive polymer composites [51], concrete [52, 53], proteins [54], etc., as shown in Figure 2. The emergence of these new inks and multi-material printing techniques has enabled DIW technology to exhibit more functional properties, thus expanding its potential applications in different fields.

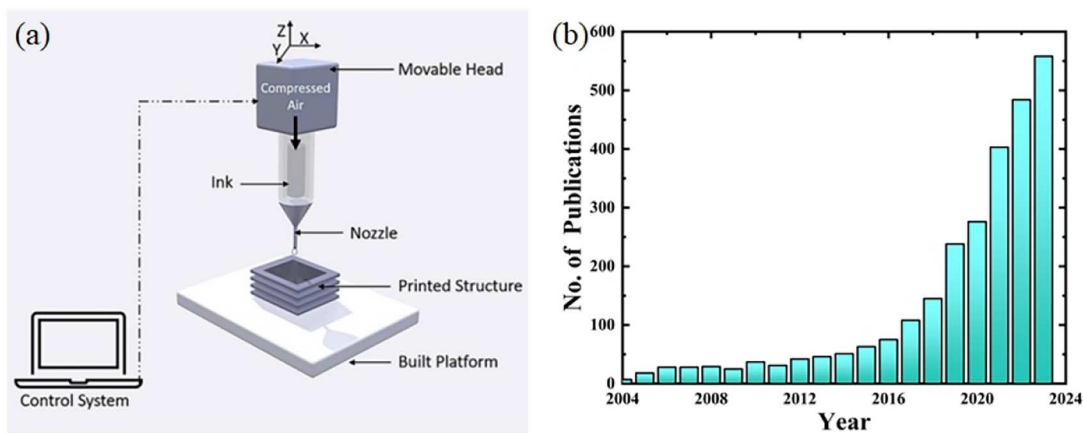


Figure 1. (a) The schematic illustration of Direct-Ink-Writing (DIW) technology [24]; (b) Recent trend in the number of publications on DIW. (© 2024, Elsevier) Data from Web of Science (December 2023, Key words: Direct ink writing).

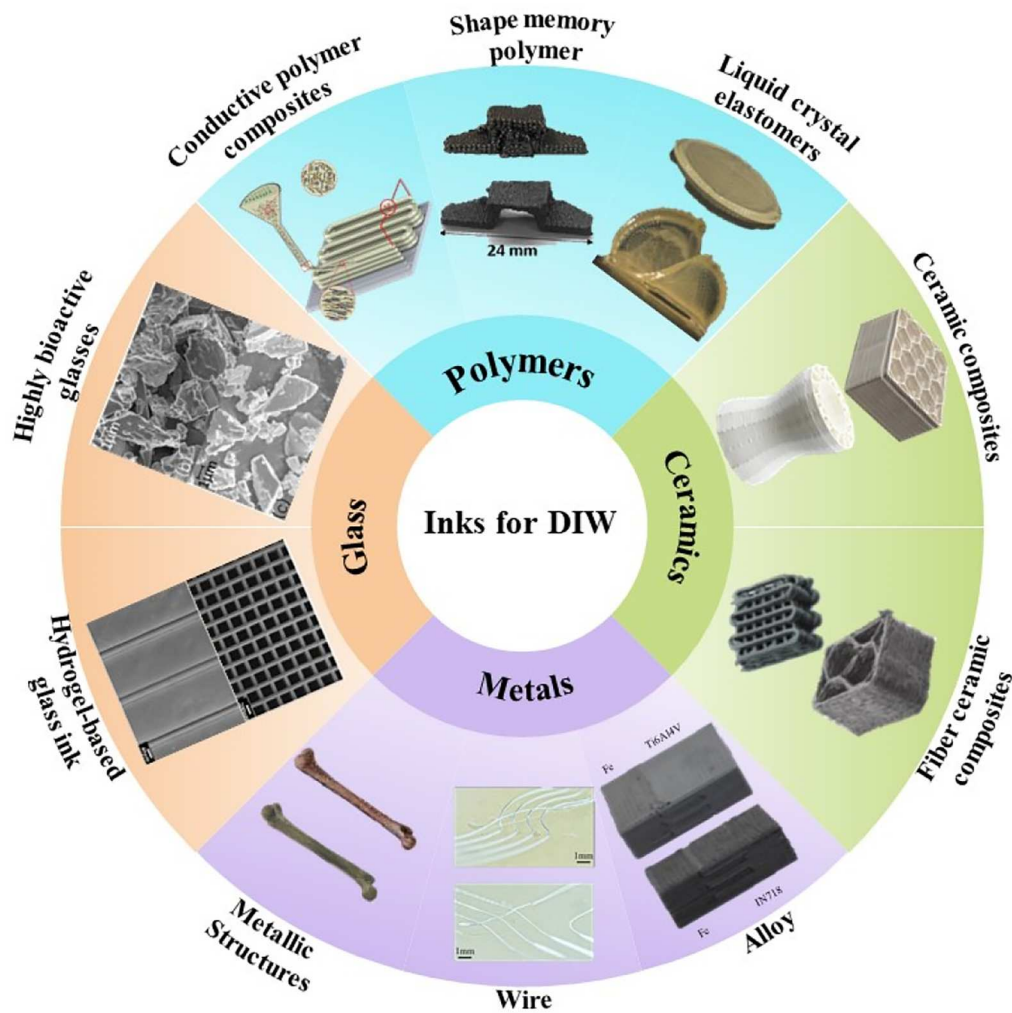


Figure 2. The inks of DIW 3D printing technology [31, 32, 34, 36, 39, 41, 43, 47, 50, 51] (© 2020, Wiley-Blackwell; © 2021, Elsevier BV; © 2019, American Chemical Society; © 2023, Elsevier Editora Ltda; © 2011, Elsevier BV; © 2018, Elsevier BV; © 2024, Wiley-VCH Verlag; © 2018, Wiley-Blackwell; © 2024, Taylor and Francis Ltd.; © 2022, Elsevier BV.).

The issue of ink in DIW technology has evolved tremendously over time. Along with different types of inks being used in DIW 3D printing technology, researchers have gradually explored the potential of the technology in various areas, including biomedicine [55–57], chemistry [58, 59], future foods or medicine [60, 61], materials science [62, 63], electronic devices [13, 64, 65], and energy storage [66–70]. In addition to the above-mentioned areas, DIW 3D printed sensors have received a great deal of attention. Researchers have successfully demonstrated the fabrication and application of a variety of sensors via DIW technology.

2.2. DIW printed sensors

In the field of sensors [5], the DIW manufacturing process is much simpler compared to other processes such as spraying [71, 72], screen printing [73, 74] and

drop casting [75, 76]. DIW technology also enables complex structural design and fabrication compared to other moulding methods. DIW-sensors can be directly printed onto the surface of the object under test, offering significant advantages over traditional adhesive methods. The improper adhesive process for traditional sensors is often susceptible to systematic errors, which may originate from misalignment, unequal adhesive thickness, and air gap discrepancies. Such errors have the potential to adversely affect the accuracy and sensitivity of the sensors. In contrast, DIW sensors can be fabricated directly on the test substrate, ensuring more precise and consistent contact with the base, thereby significantly enhancing the reliability of the sensors. In addition, advances in ink materials enable DIW sensors to be manufactured with materials selected according to the appropriate requirements to satisfy different sensor functionalities. Currently, researchers have

successfully produced a series of sensors with different functions via DIW technology, including strain sensors [77, 78], biosensors [79], flexible sensors [80–82], and temperature and humidity sensors [83–85], as shown in Figure 3.

The absence of etching, moulding, and mask-making processes in the fabrication of DIW-sensors results in a markedly streamlined sequence of steps from design to manufacturing. This confers a distinctive advantage in the expeditious fabrication of small-batch, bespoke, and intricate-shaped sensors. Accordingly, DIW sensors are particularly well-suited to the individualised requirements of sensing functions and layout schemes in the online monitoring of massive equipment. To illustrate, the most widely applied DIW strain sensors will be considered as a case. DIW strain sensors are frequently employed for the monitoring of strain in critical areas and specific locations on major equipment. The typical DIW strain sensor is composed of a multilayered composite structure comprising heterogeneous polymer materials. These include an insulating layer, a sensing layer and an encapsulation layer. The insulating and encapsulation layers are typically composed of high molecular resins, whereas the sensitive layer is usually formed of a composite material containing conductive particles (such as metal or carbon pastes). These DIW strain sensors serve to indirectly reflect the strain of

the substrate, through the means of resistance changes that are caused by the deformation of the internal sensing layer.

In comparison with traditional metal foil strain sensors, DIW strain sensors exhibit significant advantages in terms of their design, manufacturing, and placement [86]. Furthermore, current research has significantly surpassed the inherent constraints of traditional metal foil strain gauges, which are distinguished by a low gauge factor (≤ 2) and a restricted strain range ($\leq 2\%$). For example, Yu and Zhao et al. [87] achieved the high-precision fabrication of strain sensors on polyimide (PI) and aluminum alloy substrates using DIW technology. This process enhanced the sensitivity and tensile properties of the sensors, and significantly improved the measurement accuracy by introducing a coating layer. Subsequently, Guo and Yu et al. [88] proposed a new method to fabricate high-precision multilayer strain sensors based on the DIW technique. A corresponding theoretical model and finite element simulations were established to predict the nonlinear thickness dependence of the interfacial strength by optimising the DIW fabrication parameters and other processes. Ultimately, the fabricated strain sensors demonstrate high sensitivity (GF up to 28.5), a broad strain range ($>10\%$), and remarkable stability under static and cyclic tensile loading. Based on the

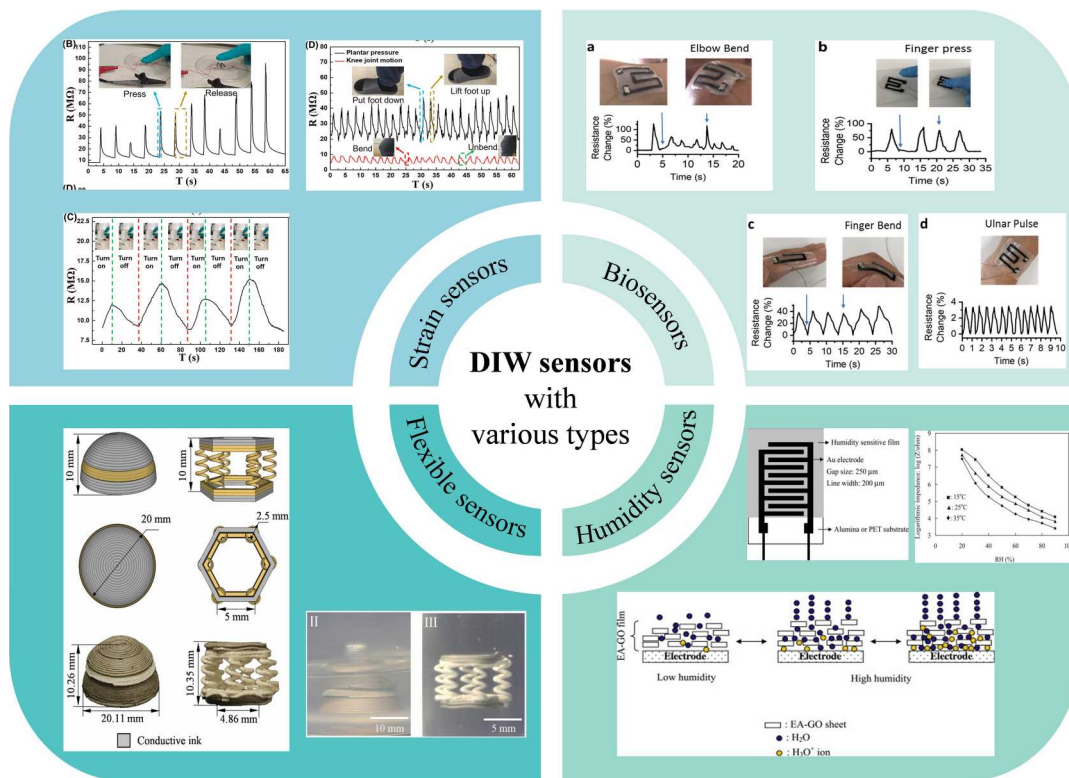


Figure 3. DIW sensors with various types [78, 79, 82, 85] (© 2023, Elsevier BV; © 2019, Elsevier BV; © 2022, Elsevier BV; © 2015, Elsevier).

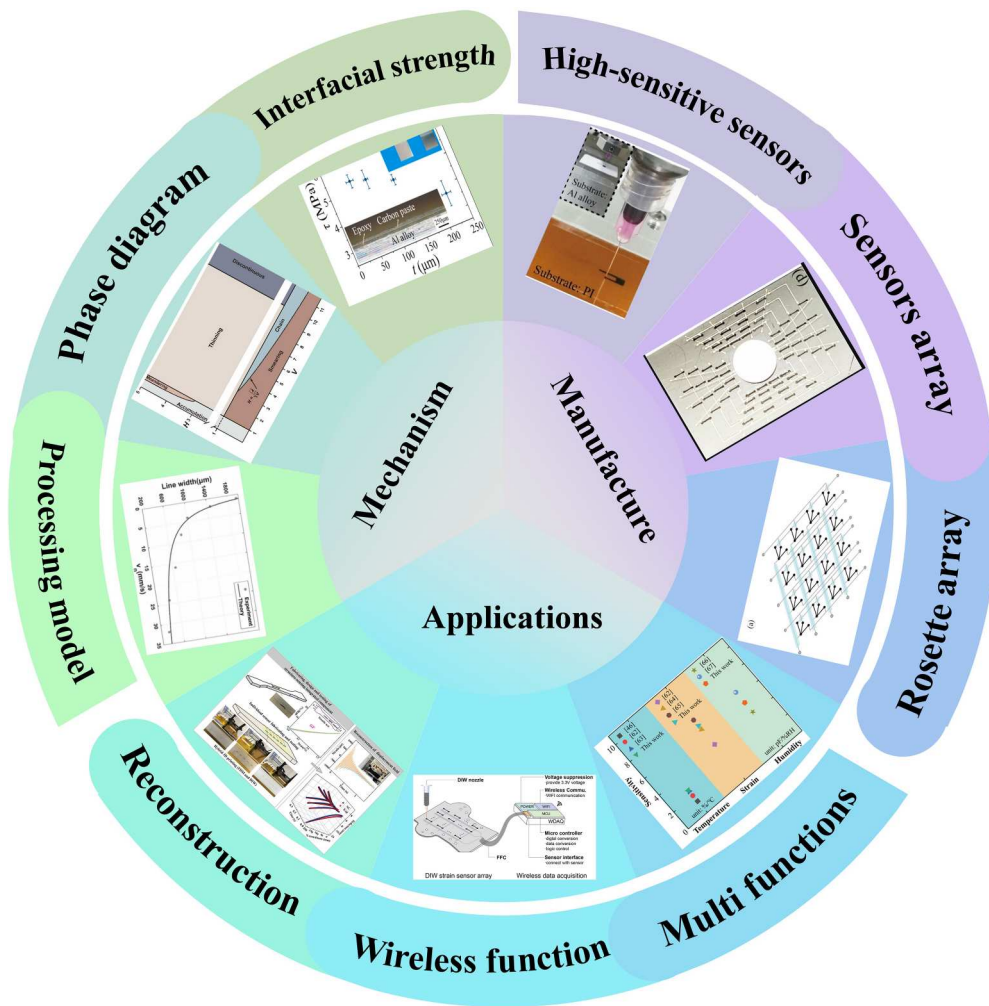


Figure 4. The development of DIW strain sensor in the mechanisms, manufacturing and applications [87, 88, 90–92, 94, 95] (© 2019, Elsevier BV; © 2021, Elsevier BV; © 2023, Elsevier BV; © 2024, Elsevier Ltd.; © 2022, Elsevier BV; © 2024, Taylor and Francis Ltd.; © 2024, Elsevier Ltd.).

aforementioned [87, 88] and subsequent process [89–91] optimisation, Yu et al. further optimised the structural and functional design of the DIW strain sensors. They then proceeded to produce sensor arrays [92], rosette-type sensors [93], and strain-temperature-humidity multifunctional sensors [94]. Additionally, they combined wireless communication technology with the DIW sensors [95], achieving wireless, real-time monitoring of object strain. Significant advancements have been made in the realms of mechanism analysis, manufacturing, and application with regard to DIW strain sensors, as illustrated in Figure 4.

3. The challenges of long-term stability and durability for DIW 3D-printed sensors

To achieve long-term and continuous online monitoring of large equipment with DIW sensors, the sensors must demonstrate both superior performance metrics as well

as long-term stability and durability that match or exceed the anticipated service life of the equipment. However, there are still relatively few reports focusing on the long-term stability and durability of DIW sensors, which has resulted in these two unresolved issues becoming significant challenges that impede the widespread adoption of DIW polymer sensors [96, 97].

To further comprehend the two key challenges that DIW sensors encounter in real-world applications, it is vital to first establish a clear distinction between long-term stability and durability. The long-term stability is used to describe the capacity of a sensor to maintain consistent performance (such as accuracy, sensitivity, response time, etc.) over an extended period. The objective is to ascertain whether the sensor is capable of retaining its original operational characteristics over an extended period without exhibiting significant drift. In contrast, the durability refers to the sensor's ability to withstand external loading factors (such as physical

shocks, fatigue loads, mechanical wear, etc.) and usage conditions, while maintaining its functionality and performance. The durability emphasises the sensor's physical resilience, which is the ability to withstand harsh conditions without being easily damaged or failure.

In essence, long-term stability is concerned with the measurement performance of the sensor, whereas durability is primarily concerned with the structural integrity of the sensor. The absence of long-term stability can lead to a significant decrease in the sensitivity of the sensor, while the absence of durability can lead to a reduced service life and sudden failure of the sensor. A comparison of the primary influencing factors, mechanisms of action, and results of these two challenges, as presented in Table 1 and Figure 5, reveals some differences.

3.1. The long-term stability challenge of DIW sensor

Long-term stability is critical to the performance of DIW sensors. Different from metal or semiconductor sensors, the DIW-printed sensors are typically fabricated by

Table 1. Overview of long-term stability and durability challenges for DIW sensors.

	Long-term Stability	Durability
Mechanisms	Ageing, deterioration	Damage, fatigue
Influential factors	Environment, manufacturing process	Loads, design of sensors
Impact on sensors	Accuracy, sensitivity, response time	Service life, structural integrity
Result	Performance degradation	Failure

extruding functional polymer pastes. As a consequence, these sensors often experience performance degradation over time induced by environmental aging or material deterioration. Currently, the existing research tends to focus only on the short-term stability of the DIW sensor. The exploration of long-term stability is still in the initial stage, and the comprehensive and in-depth study remains scarce.

DIW sensors are typically required to operate under various environmental conditions for substantial periods, including elevated temperatures, high humidity, acid and alkali corrosion, oxidation, and chemical contamination. These environmental factors have a significant impact on the long-term stability of the sensors [98, 99]. For instance, elevated temperatures [100], may induce material expansion or melting, thereby affecting the sensor's sensitivity [101]. High humidity environments, on the other hand, can lead to material absorption of water, resulting in changes in electrical properties. Chemical contaminants may react with the materials, altering their structure or performance. Furthermore, it is important to note that a range of environmental factors have the capacity to interact synergistically with the polymer materials of DIW sensors. For instance, factors such as light, oxygen, humidity and heat in the environment can collectively induce chemical changes in the polymer materials of the sensors, leading to ageing phenomena [102], as illustrated in Figure 6. Researchers have studied the ageing of different flexible substrates (e.g. polyethylene naphthalate – PEN, polyethylene terephthalate – PET)

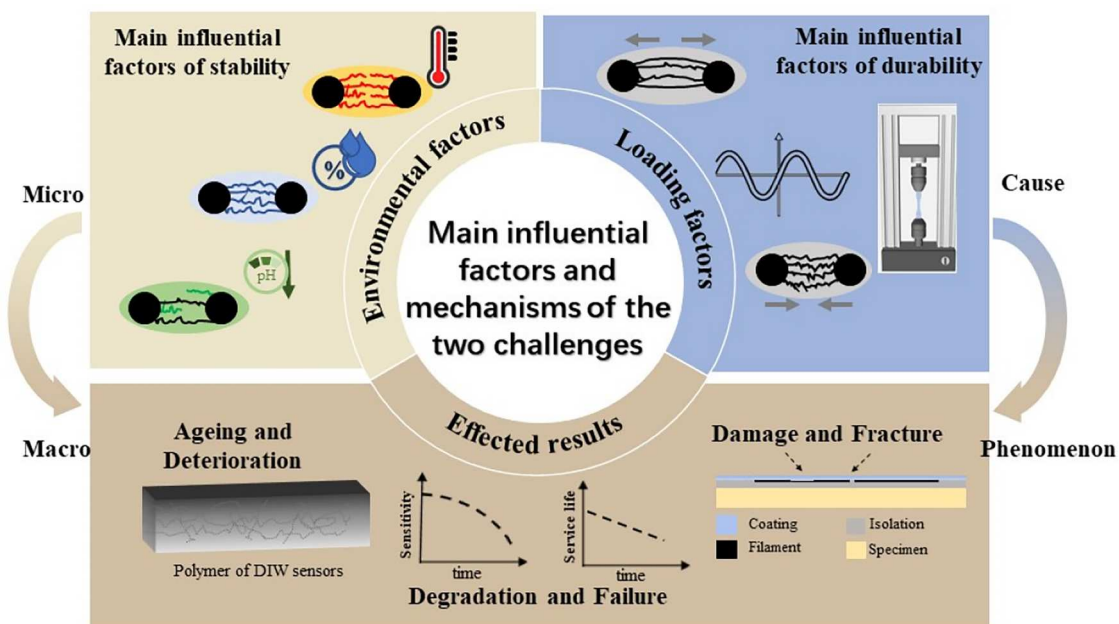


Figure 5. The main influential factors and mechanisms of the two challenges.

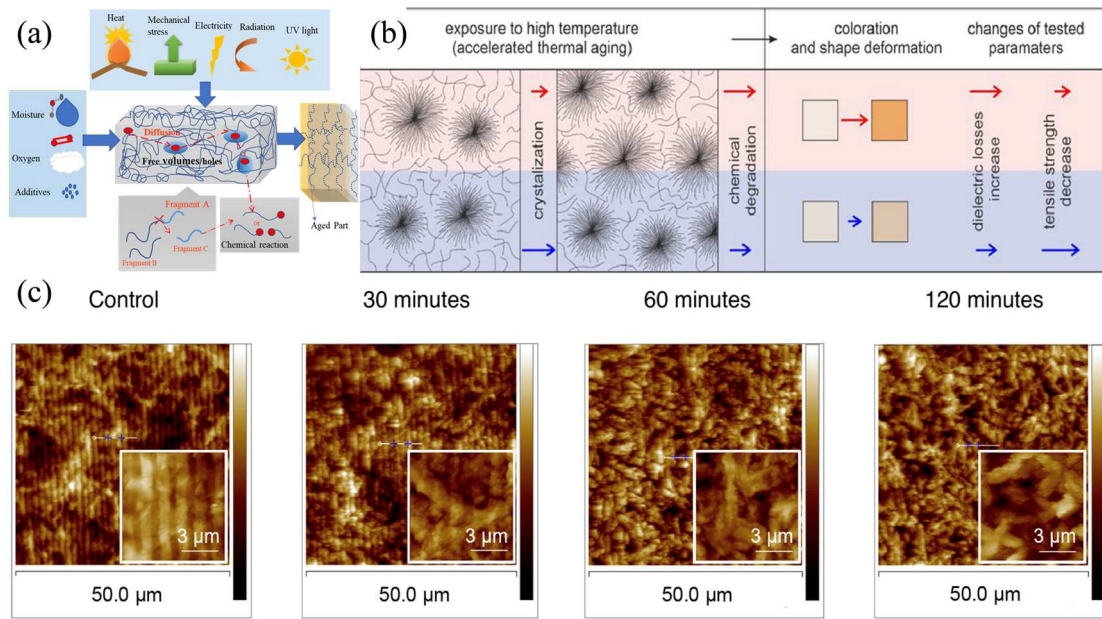


Figure 6. Aging of polymer materials: (a) Main influence factors [104]; (b) The effect of aging on materials [103]; (c) Characterisation of polymer material aging under accelerated aging test [105]. (© 2024; Elsevier Ltd; © 2022; Elsevier Ltd; © 2017; Elsevier Ltd).

and found that their ageing behaviour is basically similar. As shown in Figure 6(b), where the red line represents PEN and the blue line represents PET, the microscopic ageing behaviour is similar and both show a decrease in dielectric and tensile strength at the macroscopic level [103]. From a microscopic perspective, the aging of polymer sensors primarily involves the scission, cross-linking, and hydrolysis of polymer chains. The scission of these polymer chains reduces the length of the molecular chains, which reduces both the mechanical strength of the material and the efficiency of electrical signal transmission. Cross-linking or oxidation of polymer chains results in the covalent bonding of different molecular chains, leading to increased brittleness of the material. The hydrolysis of polymer chains, where water molecules react with the polymer chains, significantly affects the transmission of electrical signals in the polymer. Consequently, these microscopic changes often act in combination, resulting in macroscopic deterioration and performance degradation in sensors, thereby compromising long-term stability.

The long-term stability of DIW sensors is influenced by a number of factors, including environmental influences, as well as various stages within the manufacturing process. To ensure sensors maintain performance stability over extended periods, it is vital to consider the selection of the ink materials, the precision of the manufacturing process, and the post processing. These factors impact both the initial performance and the long-term stability of the sensors. For instance, if the rheological properties of the ink are not stable

enough, it may lead to defects in the microstructure during the printing process, thereby reducing the mechanical stability and electrical performance of the sensors over long-term use. It is also important to note that different printing parameters (deposition rate, nozzle movement speed, resolution, printing pressure, etc.) and post-processing techniques (sintering temperature, curing speed, drying time, etc.) can also affect the uniformity and reliability of DIW sensors. These effects can be gradually magnified over time and under the influence of the environment, ultimately leading to a decrease in the reliability of the sensing elements and premature failure of the entire system.

3.2. The durability challenge of DIW sensor

DIW sensors are often subjected to a variety of loads in long-term service, including fatigue loads, shock loads, and vibration loads, etc. Among them, fatigue loads have a major impact on the durability of sensors. Fatigue loading is a form of cyclic loading that induces periodic stress and strain within the sensor materials or structures. As illustrated in Figure 7, with an increase in the number of cycles, phenomena such as micro-cracks, dislocation accumulation, and void formation may gradually occur within the material, ultimately leading to macroscopic damage [106]. In essence, fatigue loads have a substantial and considerable impact on the long-term application of DIW sensors, significantly reducing their service life. In the domain of industrial engineering, sensors with a limited lifespan

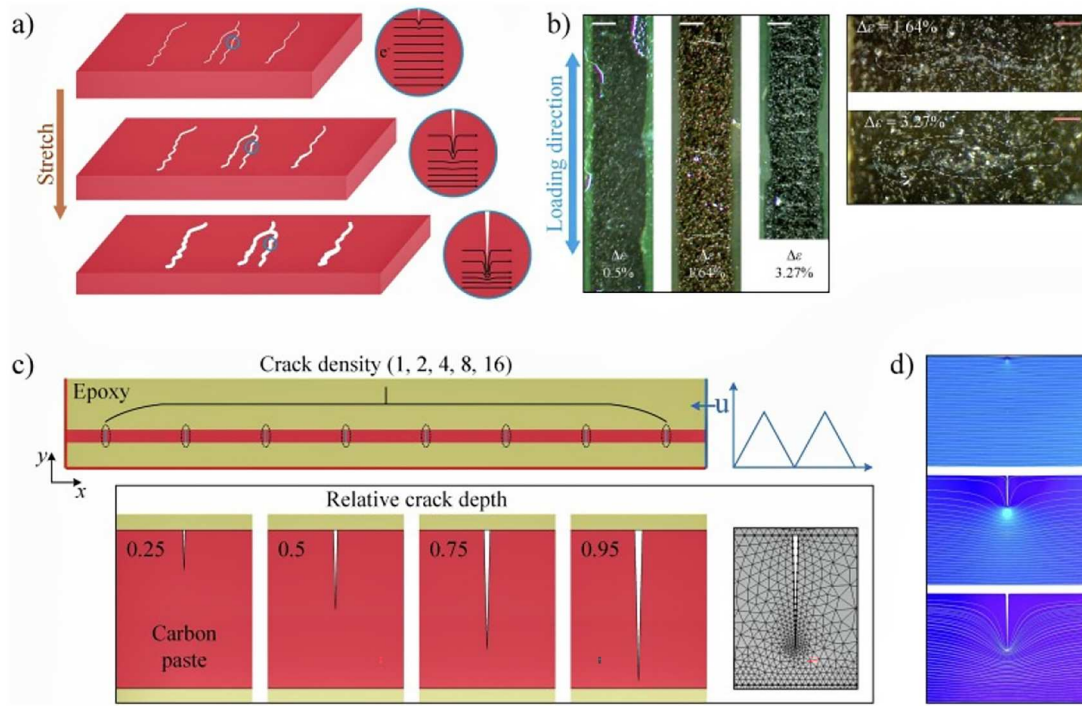


Figure 7. Fatigue characterisation and simulation of DIW sensors [107]: (a) Schematics of the widened crack size and the hindered electron transfer near the crack under stretching. (b) Optical images of the sensors in tension state, corresponding to the fatigue tests with $\Delta\epsilon$ from 0.5 to 3.27% (left, the white scale bar is 200 μm). The representative micro-crack surface morphology in carbon paste sensing layer (right, the pink scale bar is 50 μm). (c) Simplified 2D shell FE models with various crack proportion levels for the electro-mechanical coupling analysis, and (d) the distorted current density streamlines in sensing layer during stretching. (© 2023; Elsevier Ltd).

necessitate more frequent inspection, maintenance, and replacement, consequently leading to increased operational costs. Moreover, the premature failure of sensors can affect the efficiency of the system and even lead to safety issues. Consequently, to ensure the effective and widespread implementation of DIW sensors in industrial engineering, it is imperative to comprehensively understand their fatigue behaviour and accurately determine their service life.

The assessment of sensor performance with regard to long-term fatigue is of critical importance in order to ensure their durability in real-world applications. However, the lack of standardised test methods [108] has resulted in a paucity of data for predicting the durability of sensors [97]. In the course of research conducted on 3D-printed sensors, fatigue durability tests are frequently conducted. These tests are tailored to the specific sensitive materials and structures of the sensors. Some of the testing methods are shown in Table 2.

As demonstrated in Table 2, strain-controlled loading is the preferred method for durability fatigue testing of 3D-printed sensors. This is due to the fact that polymers are materials with significant viscoelastic properties, frequently exhibiting time-dependent responses (e.g.

Table 2. Fatigue test methods for some different 3D-printed sensors.

Types and material of 3D-printed sensors	Number of cycles	Load mode	Max load
Strain sensors made by double network hydrogels [109]	10	Strain	800%
Strain sensors made by epoxy nanocomposites with CNT [110]	10	Strain	8%
Strain sensors made by TPU and CNT [72]	10^3	Strain	300%
Capacitive sensors made by EIG [111]	10^3	Strain	250%
Capacitive sensors made by AgNWS /TPU [112]	10^3	Strain	30%

creep and stress relaxation) during loading. Constant stress loading would lead to significant deformation of the material over time, failing to accurately reflect the actual performance of sensors under periodic deformation conditions. Strain-controlled loading has been demonstrated to offer a more realistic simulation of the fixed deformations that sensors experience in practical applications. Furthermore, it has been shown to effectively evaluate the strain-sensitivity relationship of the sensors. Nevertheless, the selection of loading strain amplitude and frequency remains challenging when 3D-printed sensors are made from different materials. For instance, setting too low strain amplitude

may not be sufficient to activate fatigue mechanisms, while setting too high strain amplitude can lead to untypical fracture behaviour. Moreover, high-frequency loading may induce heating and thermal softening effects in polymer materials, thereby masking their actual fatigue characteristics, whereas low-frequency loading significantly extends the experimental time and increases costs. These complex mechanisms result in considerable differences in the experimental design of durability testing for 3D-printed sensors.

Furthermore, the durability tests typically conducted on recently developed novel sensing technologies are limited to $10^3 \sim 10^4$ cyclic loads [72, 113, 114], which is considerably below the threshold of 10^5 loading cycles required [115, 116] for general online monitoring engineering applications. While there have been reports in the literature of long-life (10^5 cycles) [117] or durable strain sensors that have been tested for multiple days [118], the loading levels are relatively low ($0.03\% \sim 0.2\%$), which do not accurately reflect the large loads that may be sustained in actual use. Furthermore, the maximum load applied to sensors fabricated from different materials varies significantly. For instance, as previously discussed, sensors constructed from hydrogels and certain printing inks often demonstrate strain levels surpassing 300%. Conversely, traditional conductive inks generally exhibit strain levels below 40%. The absence of definitive standards for sensor endurance testing has been identified as a primary factor contributing to substantial variations in test data across different sensors. These variations encompass both the duration of loading and the maximum load applied. These disparities and shortcomings contribute to the current inadequate comprehension of the endurance characteristics of 3D-printed sensors.

Furthermore, stress concentrations resulting from the sensor design are also a major issue affecting the durability of sensors [119–121]. These stress concentration areas are more susceptible to localised deformation or damage when subjected to external loads over long periods of time, thus reducing the service life of the sensor [122]. The issue of how to balance the design of sensor structures to address the functionality and durability of sensors still requires more in-depth and systematic research.

3.3. The synergistic effect of stability and durability

Although the characteristics of long-term stability and durability challenges affecting DIW sensors are different, the two challenges tend to be simultaneous and coupled during the service of the sensor. For example,

it is evident that environmental factors exert a considerable influence on the long-term stability of sensors. To a certain extent, these factors also affect the durability of the sensors. Environmental changes, including temperature, humidity, and exposure to corrosive elements, have been shown to accelerate material ageing and degradation, thereby undermining the long-term stability of the sensor. Concurrently, these factors can induce interlayer stresses or surface damage, precipitating structural failures that compromise the sensor's durability [123, 124]. Another example, Fatigue loading is another critical aspect, as it affects both the stability and durability of DIW sensors. It has been demonstrated that fatigue loading can shorten the service life of the sensor through the application of cyclic mechanical stress, as well as accelerating the deterioration of sensitive materials. This, in turn, has the effect of further compromising the sensor's reliability. The coupled interactions highlight the intricate relationship between stability and durability, where challenges in one domain often amplify issues in the other.

A particularly significant issue that illustrates the interaction between long-term stability and durability is the interfacial bonding within the layer-by-layer stacked DIW structures [125]. The quality of interlayer adhesion is fundamental to maintaining the structural integrity and functional reliability of DIW sensors. Over time, environmental factors such as thermal and humidity cycling [126], as well as mechanical stresses, can lead to interfacial issues including delamination, cracking, or peeling. These interfacial degradations have dual impacts: they disrupt electrical continuity, thereby compromising long-term stability, and they weaken the mechanical structure [127], thus reducing durability. This underscores the critical need to address both challenges simultaneously when optimising DIW sensors for long-term use.

To summarise, the challenges of long-term stability and durability in DIW 3D-printed sensors are interconnected, driven by environmental factors, fatigue loading, and interfacial bonding issues. These challenges encompass material aging, chemical degradation, mechanical fatigue, and interfacial failure, often acting in a coupled manner. It is therefore vital to develop a comprehensive understanding of these interactions if DIW sensors are to be developed with enhanced reliability and longevity.

4. Strategies for enhancing the long-term stability and durability of DIW sensors

Long-term stability and durability challenges are still the main obstacles limiting the prolonged and widespread use of DIW 3D-printed sensors. The two challenges

often combine and interact simultaneously. Therefore, when adopting appropriate strategies to improve the long-term stability and durability of sensors, the combination of the two challenges should not be divided, but should be considered holistically. In addition, the environmental influences and fatigue loads on sensors in service are not controllable, but rather random. Accordingly, we focus on the manufacturing aspects of the sensors to provide some potential strategies to improve the long-term stability and durability of DIW sensors, including ink innovations, design for the structure of sensors and optimisation of printing process, as shown in Figure 8.

4.1. Innovations of inks

The selection and optimisation of ink materials are important in the enhancement of the stability and durability of DIW 3D-printed sensors. Printing inks for DIW technology must fulfil the requirements of the manufacturing process, and have excellent mechanical and electrical properties, as well as environmental adaptability [128]. The optimisation of material formulation is a central part of this process [129]. By changing the chemical composition of the material and adding intensifiers or modifiers, the mechanical strength, electrical conductivity and environmental resistance of the material can be significantly improved.

For example, Zhao et al. [130] using multi-walled carbon nanotube-enhanced nanocomposites in fibre-glass reinforced composites for potential damage diagnostics and structural health monitor applications. The

objective of the study is to enhance DIW printing by optimising the ink. Wang and Xu et al. [131] proposed a SiCN/RuO₂/TiB₂ composite ceramic ink that can be used for DIW printing. Based on this ink, temperature sensors and heat flux sensors with high linearity and high stability (resistance drift rate of 0.1%/h at 800 °C) were successfully prepared. The stability and durability of DIW sensors was enhanced by optimising the ink material. In addition, Kaplan and Ozkul [132] previously proposed a design concept for a smart damage sensor for monitoring structural fatigue. They considered that the fabrication of the smart damage sensor would require the specific material which needed to determine the quantitative relationship between the number of cycles and the damage to the material. However, existing DIW inks are not yet able to satisfy this requirement. This demonstrates the importance of ink optimisation for sensor design and fabrication.

Moreover, the development of multifunctional composites to enhance the long-term stability and durability of sensors has become an important area of research [42, 133–135]. The long-term stability and durability of DIW sensors can also be successfully enhanced [136] by combining conductive materials with high-performance polymers [137] or by introducing self-healing [134] components into the conductive paste.

4.2. Design for structures of sensors

The geometric configuration of the DIW sensor constitutes the fundamental basis of its structural design, exerting a substantial influence on its mechanical behaviour.

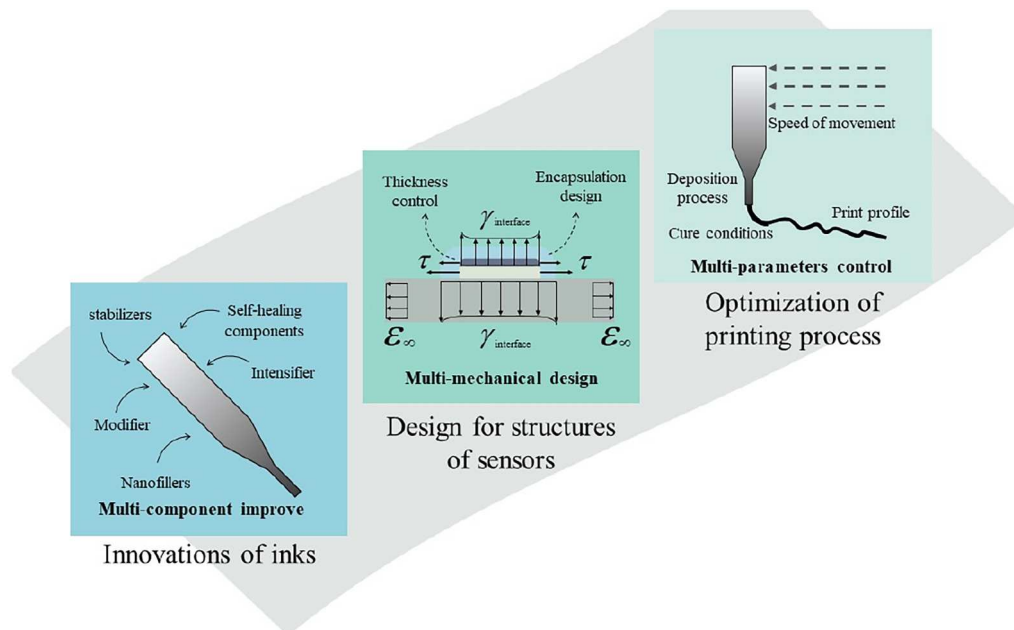


Figure 8. Strategies for enhancing the long-term stability and durability of DIW sensors.

In the context of the core sensing layer, the design must fulfil the operational requirements of the sensing element, whilst also undergoing geometric optimisation to enhance long-term stability and durability. The purpose of optimising the geometry is to reduce the occurrence of stress concentration and mechanical fatigue, thus improving the durability of the sensor. For example, optimised geometries can be used in the design to reduce stress concentration areas, such as rounded corners, smooth transitions and avoiding sharp corners [138, 139], as shown in Figure 9 (a).

Building on the foundational geometry of sensing layer, the protective encapsulation design plays a crucial role in safeguarding the sensor from environmental challenges while preserving its functionality, which also enhances the long-term stability and durability of sensors. The encapsulation design involves not only the selection of materials, but also the optimisation of the encapsulation, and these factors combined determine the performance and reliability of the sensor in practical applications. In material selection, materials with excellent mechanical properties and chemical stability should be chosen to ensure that the encapsulation can maintain its structural integrity under various environmental conditions. In addition, the design of the encapsulation needs to consider the strain transfer efficiency of the sensor, as shown in Figure 9 (b), i.e.

how to protect the sensor while ensuring that the strain signal can be effectively transferred to the sensor's sensitive element. Advanced encapsulation designs, such as those employing multi-layered or composite structures, further improve durability and adaptability.

In addition to geometry and encapsulation, the strength of interlayer interfaces within the sensor is a critical factor influencing its overall structural stability and long-term sustainability. The selection of appropriate materials plays a key role in improving the adhesive strength of the interface and enhancing the adaptability of the sensor to changes in the external environment. Equally important is the control of layer thickness, which can optimise the stress distribution within the material and reduce interlayer stress concentration, as shown in Figure 9 (c), thereby improving the sensor's durability. Furthermore, advancements in interlayer interface processing technologies contribute significantly to strengthening the bonding. Techniques such as optimising printing parameters and applying specialised surface treatments can enhance interlayer adhesion and improve the overall structural stability of the sensor [140, 141].

The geometric configuration, protective encapsulation, and interlayer interface strength collectively form an integrated framework that determines the overall

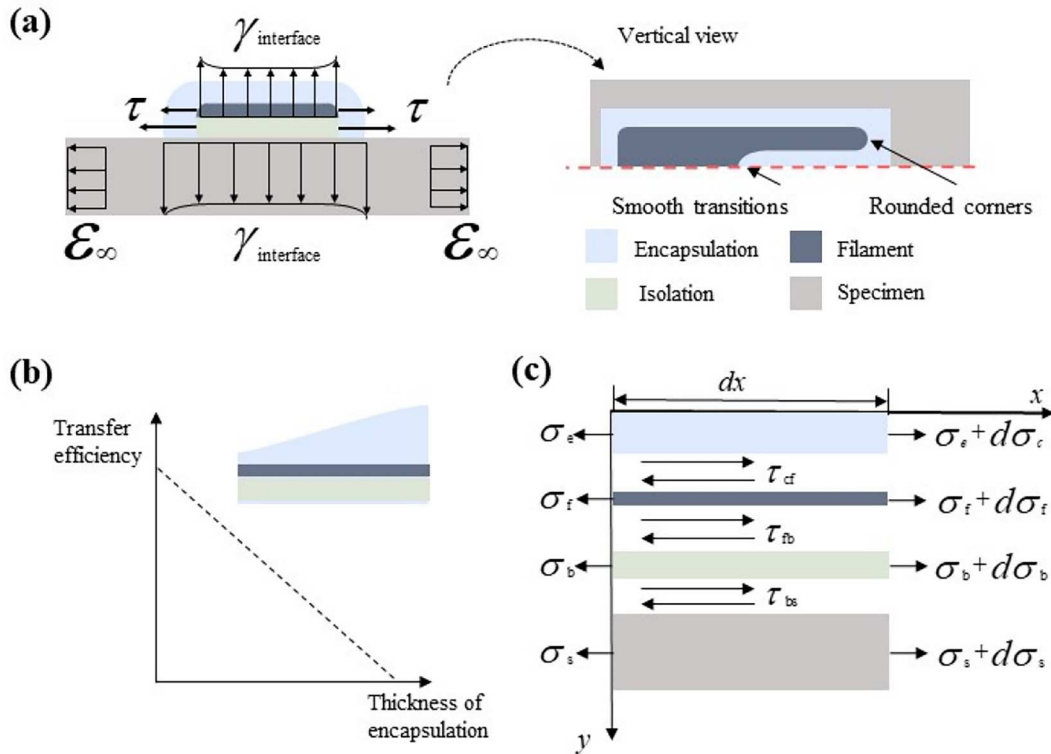


Figure 9. (a) Mechanical design of DIW-sensors with smooth transitions; (b) Illustration of the DIW sensor transfer efficiency versus thickness of encapsulation; (c) Layer-thickness design and stress distribution of DIW-sensors.

performance and reliability of DIW sensors. Each design element contributes uniquely to enhancing the sensor's durability, with geometry optimisation addressing stress concentration, encapsulation safeguarding against environmental challenges, and interlayer bonding ensuring structural integrity. By harmonising these aspects, it becomes possible to develop sensors capable of withstanding diverse operational demands while maintaining long-term stability and functionality. This comprehensive approach underscores the importance of a holistic design strategy in advancing the capabilities of DIW-printed sensors.

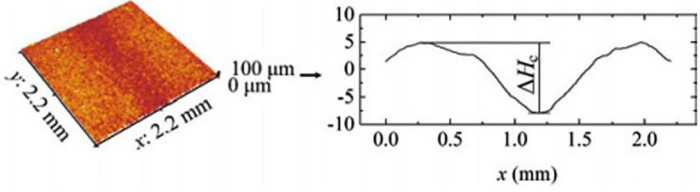
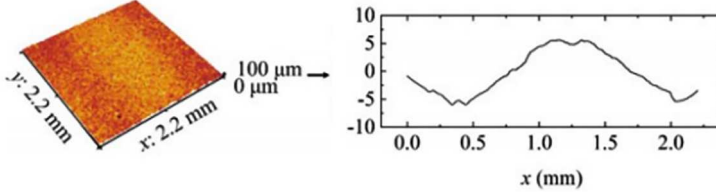
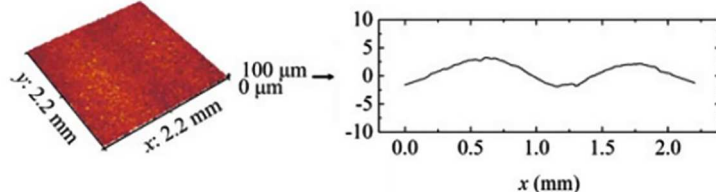
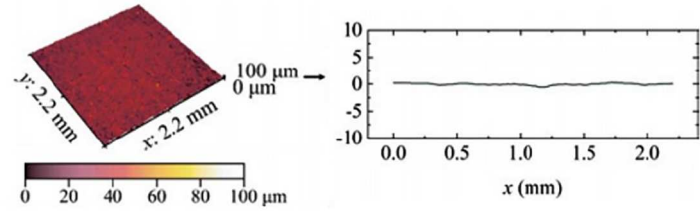
Table 3. Optimisation of the DIW fabrication parameters and corresponding flatnesses for epoxy layers [88]. (© 2021, Elsevier BV).

	D_{needle} (μm)	S_{spacing} (μm)	P_{air} (psi)	ΔH_e (μm)
1	700	950	15.5	16.89
2	700	800	10.5	14.46
3	700	650	10.5	7.78
4	250	150	25.5	2.43

4.3. Optimisation of printing process

One of the strategies to improve the long-term stability and durability of DIW sensors is to ensure the uniformity and consistency of the sensor monolayer. To achieve this, a number of parameters need to be synergistically controlled, including, but not limited to, the rheological properties of the ink [142], the nozzle travelling speed [90, 143], the deposition process [144], the resolution [145], and the curing conditions [146, 147]. The slight differences in these parameters may affect the final printing quality and affect the internal-stress distribution of the sensor, which could ultimately reduce sensor life. For example, Tables 3 and 4 [88] illustrate the flatness parameters and morphologies of epoxy under different printing parameters. In the tables, D_{needle} represents the inner diameter of the printing nozzle, S_{spacing} represents the printing line spacing, P_{air} represents the printing air pressure, ΔH_e represents the maximum peak-to-valley height difference, and ΔH_c represents the

Table 4. Flatnesses of carbon paste layers printed on the epoxy layers shown in the row 1–4 of Table 3 [88]. (© 2021, Elsevier BV).

	Surface morphology	ΔH_c (μm)
1		12.89
2		11.56
3		5.12
4		0.84

maximum peak-to-valley height difference of the surface waviness morphology under white light interferometry. A printing speed of 10 mm/s was uniformly selected for the experiments. It is evident from the tables that printing parameters have a significant impact on the morphology of DIW printing. By rationally regulating the printing parameters, the accuracy of the printing process could be improved, manufacturing defects could be reduced, and the electrical properties and mechanical stability of the sensors could be enhanced [148].

The quality of DIW printing is influenced by a combination of parameters. For instance, a high printing speed generally necessitates a higher extrusion pressure to ensure uninterrupted supply of material, thereby averting material flow insufficiency that could result in short lines or damage. Conversely, a larger nozzle diameter has been shown to reduce extrusion resistance, thereby facilitating a continuous material supply at a constant pressure. However, this may potentially compromise printing resolution, resulting in an unduly thick stacking. How to efficiently regulate the printing multi-parameters has become an important issue. Based on this, we can introduce other technologies such as machine learning, multi-parameter cooperative

control systems, real-time monitoring systems, etc. to enhance the effectiveness. Additionally, the application of reliability engineering principles, such as failure mode and effect analysis (FMEA), can help designers identify potential points of failure of sensors in design and take preventive measures to improve the overall stability and durability of sensors.

In short, process optimisation is still necessary to improve the long-term stability and durability of the sensors. Process optimisation not only ensures the homogeneity of the sensor and reduces the generation of internal defects, but also improves the sensitivity and functionality of the sensor. However, how to synergistically optimise the multi-parameters of the printing process still needs to be explored in more depth.

5. Prospects of future studies

Although there are a few potential strategies available, they still cannot fully overcome all the challenges. Therefore, in-depth and innovative research is still required for DIW sensors to have long-term reliable performance in a wider range of applications. The following sections outline the crucial areas, as shown in Figure 10, where future research should focus in order to overcome

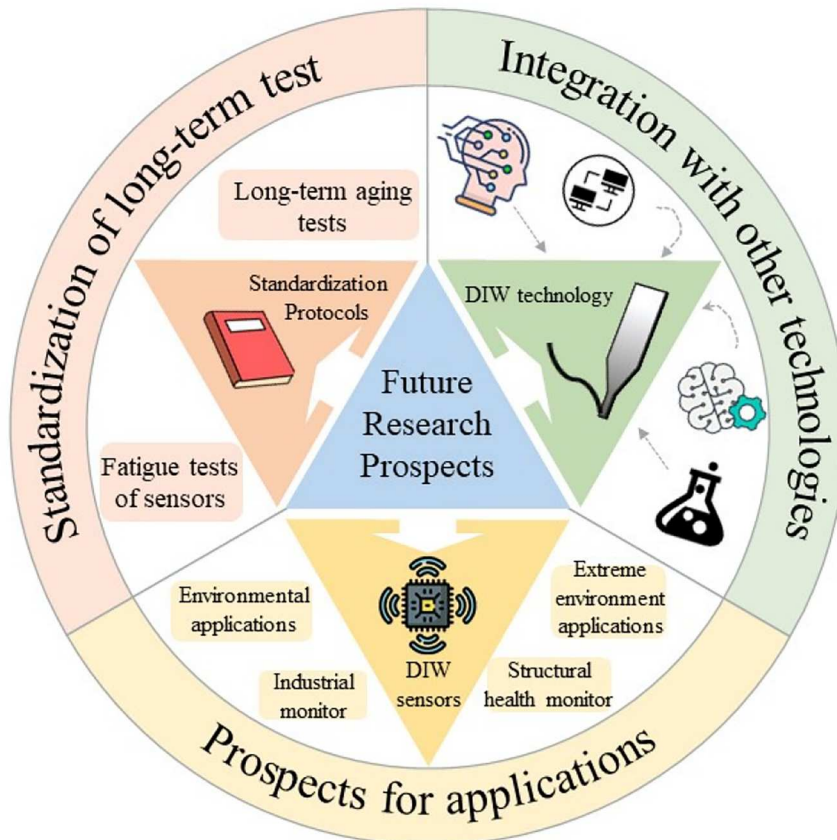


Figure 10. Prospects of future studies for the DIW-printed sensors.

these persistent challenges and unlock the full potential of DIW technology in practical applications.

5.1. Standardisation of testing protocols for long-term performance

While the potential strategies discussed (e.g. process optimisation and structural design improvements) provide a basis for improving the long-term stability and durability of DIW sensors, these strategies must be rigorously validated [149]. It is not only necessary to strengthen the mechanical theory [150], but also need to carry out a large number of experiments [107, 151] or simulations [152]. However, current test methods for DIW sensors are still fragmented and lack standardised protocols, making it difficult to systematically assess their long-term performance. For instance, as previously mentioned, current research on the fatigue performance of sensors is often tailored to the specific sensor structures and materials being designed. There are considerable differences among fatigue testing methods for different sensors, both in terms of cycle counts and load levels. These discrepancies in methodologies give rise to challenges in the comparison of fatigue data for sensors, thus impeding the translation of research findings into practical applications.

To further illustrate the fragmentation and insufficiencies of existing fatigue testing protocols for sensor materials, we have compiled a table of some common fatigue testing standards, as shown in Table 5.

The aforementioned standards are principally directed towards the common materials employed in sensors, and are frequently utilised to assess and evaluate the long-term stability and durability of sensors fabricated from various materials. However, it should be noted that these standards can only be partially applied to the long-term performance evaluation of sensors. The long-term performance of sensors is not solely dependent on their material properties but is

also closely related to their design structure, manufacturing processes, and operating environment.

The primary concern at present is the establishment of unified standard guidelines for the long-term performance of DIW sensors. However, in formulating these unified testing standards, it is essential to consider a multitude of factors, including the working principles, material properties, usage environment, and intended applications of different DIW sensors. For instance, the maximum load levels, load ranges, cycle counts, and loading speeds should be designed based on the different sensitive materials; the durability changes of the sensors should be quantitatively analysed based on key indicators such as sensitivity coefficient, response time, and accuracy. Furthermore, multi-physical fields, including elevated temperatures, humidity, and chemical exposure, should be established in accordance with the designated service environment of the sensors, thereby facilitating a comprehensive evaluation of their long-term performance. In the context of environment-accelerated fatigue tests, it is imperative to consider the corresponding loading ratios between environmental factors and fatigue loads [153]. This ensures that environmental factors and fatigue loads act equally and concurrently on the sensors, thereby simulating the service conditions of sensors in real environments. For sensors with different structures, standard specifications should be tailored and adjusted according to the material structures. For instance, the microstructure of sensors, such as Micro-Electro-Mechanical Systems (MEMS) sensors, necessitates special consideration of microscale size effects and surface defects. In the case of sensors with layered structures, the adhesion strength between layers, interfacial cracking, and delamination issues are of particular significance during the fatigue process. This necessitates the design of specific testing methods targeted at interlayer fatigue.

The development of these test standards is a high priority for the field, as it will provide the necessary framework for evaluating and improving the durability and reliability of DIW sensors. This will enable them to meet the stringent requirements of industrial and other demanding applications, thereby extending their availability and long-term functionality. Without these standardised tests, it will remain a challenge to fully understand the failure mechanisms of DIW sensors and improve their design and material composition accordingly.

5.2. Integration with other technologies

The fabrication process and structural design of sensors are still crucial for their long-term stability and durability.

Table 5. Summary of fatigue testing standards for different materials.

Standard number	Standardised content
ASTM D3479	Standard Test Method for Tension-Tension Fatigue of Polymer Matrix Composite Materials
ASTM D7791	Standard Test Method for Uniaxial Fatigue Properties of Plastics
ASTM E606	Standard Test Method for Strain-Controlled Fatigue Testing
ASTM D4482	Standard Test Method for Rubber Property – Extension Cycling Fatigue
ISO 12111	Metallic materials – Fatigue testing – Strain-controlled thermomechanical fatigue testing method
GB/T 43896–2024	Metallic materials – Very high cycle fatigue – Ultrasonic fatigue test method

Integration with other technologies in the fabrication and design of sensors, therefore, not only improves the various performances of sensors, but may even lead to the development of sensors with new functions.

In the fabrication process of sensors, multi-parameter cooperative control tends to determine the moulding quality and long-term performance of sensors. Integrating DIW technology with other technologies can effectively improve the manufacturing accuracy of sensors. For example, real-time monitoring and dynamic adjustment of the printing process can be realised by integrating an intelligent control system (ICS), which can improve the precision and reliability of printing [154]. In addition, integrating DIW technology with Artificial Intelligence (AI) [155, 156] may provide a rational structural design, and present precise printing parameters. For instance, Chen et al. utilised machine learning to analyse healthcare data from patients and predict the most suitable printing designs, reducing manual intervention and enhancing the accuracy and functionality of the models, thereby highlighting the potential of combining AI with 3D printing to achieve personalised medicine [157]. Kim et al. investigated how to maximise the capacitance of 3D-printed supercapacitors through a convolutional neural network (CNN)-guided Bayesian optimisation approach. By integrating machine learning models with 3D printing technology, researchers were able to optimise the distribution of materials, structure, and electrochemical performance during the design process to achieve the best energy storage effects [158]. Qing et al., on the other hand, used AI technology to analyse and predict the health status of composite materials and improved the durability of the materials by optimising the design [159]. Furthermore, DIW technology could also interacted with other technologies including electronic engineering (EE) [160], computer science [161, 162], chemical engineering, and beyond. Integration with other subjects and technologies could improve the accuracy of prototyping and manufacturing for the sensors, also the long-term stability and durability for the sensors.

In summary, DIW technology provides great potential to be integrated with other technologies. Future research can further improve the long-term stability and durability of DIW sensors through multidisciplinary cooperation, which could create more avenues for scientific research and industrial applications.

5.3. Prospects for applications

The different applications in the future will not only advance the technology, but also validate and promote the improvement of long-term stability and

durability of the sensors in various scenarios. First of all, for applications in conventional environment, the design and fabrication of sensors should primarily focus on factors such as air, water vapour, lighting, temperature changing, etc. Secondly, for industrial process monitoring, including the chemical, pharmaceutical and food processing industries, the DIW-printed sensors are required to maintain their performance under complex chemical environments and temperature variations. Subsequently, DIW sensors will be also expected to play an important role in structural health monitoring in critical areas such as bridges and aircrafts, where the sensors are suffering prolonged mechanical loads and severe in-service environments. Eventually, applications for extreme environments such as space exploration and deep ocean exploration, the DIW-printed sensors are required to withstand extreme temperature changes, high pressures, vacuum environments, electromagnetic radiation and chemical corrosion, thereby testing and enhancing their long-term stability and durability to the fullest extent.

Nevertheless, much more in-depth studies are still required to enable DIW sensors for flexible and long-term applications in complex serve environments. It is expected that the application potential of DIW sensors will eventually be fulfilled as long as the long-term stability and durability challenges of DIW technology are overcome.

6. Conclusion

This review discussed the challenges, solution strategies and future development perspectives of DIW 3D-printed sensors. DIW technology offers superior flexibility for creating sensors with complex geometries and customised functionalities, making it widely used in the sensor field. However, long-term stability and durability challenges are still significant limitations that restrict the widespread applications of DIW sensors in complex environments.

The challenges of long-term stability and durability for DIW sensors are affected by a variety of factors, including environmental factors, fatigue loads, fabrication processes, and the design of sensors. Moreover, the two challenges often arise simultaneously and synergistically. Tackling these challenges requires a multifaceted strategy that focuses on ink innovation, the improvement of design for sensor structure and the optimisation of printing process. In addition, future research direction requires not only the establishment of fatigue and accelerated aging test standards for DIW sensors, but also the integration of DIW technology with other technologies. It is expected that the

challenges faced by DIW sensors could be gradually overcome through the design-manufacture-test-optimisation process.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

We gratefully acknowledge the financial support from the National Natural Science Foundation of China [grant numbers 12372078, 12372109], National Key R&D Program of China [grant number 2023YFB4605101], Postgraduate Research & Practice Innovation Program of Jiangsu Province [grant number KYCX24_2506, KYCX24_2508], the Sixth Phase of Jiangsu Province '333 High Level Talent Training Project' Second Level Talents, and the State Key Laboratory of Mechanics and Control of Mechanical Structures Nanjing University of Aeronautics and Astronautics [grant number MCMS-E-0422G04].

Data availability statement

The authors confirm that the data supporting the findings of this study are within the article.

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